

5(a)	1 family history of, breast / ovarian, cancer ; <i>plus any three from:</i> 2 (mutation causes) change of primary structure (of tumour suppressor protein) ; 3 changes, folding / bonding / 3D shape / tertiary structure (of tumour suppressor protein) ; 4 non-functional (tumour suppressor) protein formed ; 5 DNA is not repaired / cells with damaged DNA do not die ; 6 (so) increases number of mutations / mutations accumulate ; 7 (some mutations) cause uncontrollable, cell division / mitosis (of breast cells) ;	4
5(b)	1 only small amount of DNA available ; A gene 2 amplify, DNA / (<i>BRCA1</i>) gene ;	2
5(c)	6.1×10^4 ;; A 6.125×10^4 <i>for two marks</i> <i>allow one mark for</i> 61 250 or $\frac{3.5 \times 10^7}{400} = 8.75 \times 10^4$ or 87 500	2

Question	Answer	Marks																							
5(d)	any two from: <i>supports:</i> 1 (women with <i>BCRA1</i>) have a higher <u>probability</u> of survival ; ora 2 (for women with <i>BRCA1</i>) the probability of survival decreases less or there is a smaller rate of decrease of probability of survival or bigger difference between the two groups over time for the probability of survival ; ora 3 data quote to show differences between the survival of the two groups at a particular time ; <table border="1"> <thead> <tr> <th rowspan="2">time / months</th><th colspan="2">probability of survival</th></tr> <tr> <th>with mutation</th><th>without mutation</th></tr> </thead> <tbody> <tr> <td>30</td><td>0.975 range 0.975 – 0.988</td><td>0.85</td></tr> <tr> <td>60</td><td>0.95</td><td>0.75</td></tr> <tr> <td>90</td><td>0.85</td><td>0.675 range 0.675 – 0.688</td></tr> <tr> <td>120</td><td>0.75</td><td>0.63 range 0.613 – 0.625</td></tr> <tr> <td>150</td><td>0.75</td><td>0.55</td></tr> <tr> <td>180</td><td>0.7 / 0.70 range 0.7 – 0.71</td><td>0.475 / 0.474 / 0.47 range 0.46 – 0.475</td></tr> </tbody> </table> 4 study carried out for, a long time / 15 years ; <i>does not support:</i> 5 no statistical test carried out / <i>ref. to</i> sample size ;	time / months	probability of survival		with mutation	without mutation	30	0.975 range 0.975 – 0.988	0.85	60	0.95	0.75	90	0.85	0.675 range 0.675 – 0.688	120	0.75	0.63 range 0.613 – 0.625	150	0.75	0.55	180	0.7 / 0.70 range 0.7 – 0.71	0.475 / 0.474 / 0.47 range 0.46 – 0.475	2
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